Customer Shopping Behavior Analysis

Revised Project Report Summary

Tools Used	Python • PostgreSQL • SQL • Power BI
Dataset	3,900 customer purchase records • 18 attributes
Focus	Customer behavior, purchasing patterns, product preferences, subscriptions, and revenue

This revised version preserves the project's original data, methodology, findings, and recommendations while presenting them in a more polished academic format.

1. Project Overview

The Customer Shopping Behavior Analysis project focuses on understanding purchasing patterns and customer preferences through the analysis of transactional shopping data. The dataset contains information from 3,900 customer purchases covering multiple product categories.

The primary objective is to transform raw customer transaction data into meaningful business insights. The analysis examines customer demographics, purchasing habits, product preferences, discount usage, subscription behavior, customer loyalty, and revenue patterns. These findings can support decisions related to marketing, customer retention, product promotion, and sales strategies.

2. Dataset Description

The dataset consists of 3,900 records and 18 attributes. It contains customer information, transaction details, and behavioral characteristics.

Area	Variables
Customer Information	Age, Gender, Location, Subscription Status
Purchase Information	Item Purchased, Category, Purchase Amount, Season, Size, Color
Shopping Behavior	Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating,
Missing Data	37 values in Review Rating

3. Data Preparation and Exploratory Analysis Using Python

The first stage involved preparing and examining the dataset using Python, with Pandas used for data manipulation and exploratory analysis.

Data Loading and Initial Examination: The dataset was imported using Pandas. Functions such as df.info() and df.describe() were used to understand its structure, data types, distributions, and summary statistics.

Missing Value Treatment: The dataset was checked for incomplete records. Missing values in the Review Rating field were replaced with the median rating of the respective product category.

Column Standardization: Column names were converted into snake_case format to improve readability and simplify further analysis.

Feature Engineering: Two additional variables were created: age_group, which groups customers by age, and purchase_frequency_days, derived from purchase information.

Data Consistency: The relationship between discount_applied and promo_code_used was examined. Since the fields were considered redundant, promo_code_used was removed.

Database Integration: The cleaned Pandas DataFrame was connected to PostgreSQL and loaded into the database for SQL-based business analysis.

4. Business Analysis Using SQL

The cleaned dataset was analyzed in PostgreSQL to answer practical business questions and identify patterns that could support decision-making.

Revenue Analysis by Gender: Revenue generated by male and female customers was compared. The analysis showed that male customers generated higher total revenue in the dataset.

High-Spending Discount Users: Customers who used discounts were examined to identify those whose purchase values were above the average purchase amount.

Highest-Rated Products: Products were ranked by average review rating. Gloves, Sandals, Boots, Hat, and Skirt were among the top-rated products, with Gloves recording the highest average rating of 3.86.

Shipping Method Comparison: Average purchase value was compared across shipping options. Standard shipping recorded approximately 58.46, while Express shipping recorded approximately 60.48.

Subscribers and Non-Subscribers: Spending and revenue were compared according to subscription status. Subscribers had an average spend of 59.49, while non-subscribers had an average spend of 59.87.

Discount-Dependent Products: The five products with the highest discount rates were Hat (50.00%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), and Pants (47.37%).

Customer Segmentation: Customers were classified as New, Returning, or Loyal based on purchase history. The resulting groups contained 83 New, 701 Returning, and 3,116 Loyal customers.

Top Products by Category: The analysis identified the three most purchased products within each category, helping highlight product-level demand patterns.

Repeat Buyers and Subscriptions: Customers with more than five purchases were analyzed in relation to their subscription status to examine the connection between repeat purchasing and subscriptions.

Revenue by Age Group: Revenue was compared across age groups. Young Adults contributed 62,143, Middle-aged customers 59,197, Adults 55,978, and Seniors 55,763.

5. Power BI Dashboard

An interactive dashboard was developed using Microsoft Power BI to present the analytical findings in a visual and user-friendly form.

The dashboard includes key indicators and visualizations covering total customers, average purchase amount, average review rating, subscription distribution, revenue and sales by category, and revenue and sales by age group.

Interactive filters for subscription status, gender, product category, and shipping type allow users to explore the dataset from multiple perspectives.

6. Key Business Insights

The analysis produced several useful observations about customer and product behavior.

Male customers contributed a higher share of total revenue in the analyzed dataset.

Express shipping was associated with a slightly higher average purchase amount than Standard shipping.

Gloves recorded the highest average rating among the top-rated products.

A substantial portion of customers belonged to the Loyal segment.

Young Adults generated the highest revenue among the analyzed age groups.

Several products demonstrated a strong dependence on discounts, highlighting the need for careful promotional planning.

Product preferences varied across categories, creating opportunities for category-specific campaigns.

7. Business Recommendations

Strengthen the Subscription Program: Promote subscription plans through meaningful benefits such as exclusive offers, improved services, or customer-specific rewards.

Develop Customer Loyalty Initiatives: Introduce loyalty rewards and retention campaigns to maintain engagement among loyal customers and encourage returning customers to increase purchase frequency.

Optimize Discount Strategies: Apply discounts strategically, particularly for products with high discount dependence, to increase sales while maintaining control over profit margins.

Promote High-Performing Products: Give greater visibility to highly rated and frequently purchased products through campaigns, recommendations, and product displays.

Implement Age-Based Marketing: Since Young Adults contribute the highest revenue, campaigns can be tailored toward this high-performing segment while retaining appropriate strategies for other age groups.

Use Shipping Preferences Strategically: Since Express shipping showed a slightly higher average purchase value, businesses can further evaluate whether faster delivery options influence purchasing behavior.

8. Conclusion

The Customer Shopping Behavior Analysis project demonstrates how transactional data can be transformed into actionable business insights through a structured analytics workflow.

The project combines Python for data preparation and exploratory analysis, PostgreSQL for structured SQL analysis, and Power BI for interactive visualization. Through customer segmentation, revenue analysis, product evaluation, discount analysis, subscription comparison, and age-group analysis, the project provides a comprehensive view of purchasing behavior.

The resulting insights can support improvements in customer retention, subscription adoption, promotional strategies, product positioning, and targeted marketing. Overall, the project demonstrates the value of combining data preparation, SQL analytics, and business intelligence tools to support data-driven decision-making.

Project Snapshot

Metric	Value
Records analyzed	3,900
Dataset attributes	18
Missing Review Rating values	37

Highest average product rating	Gloves - 3.86
Highest revenue age group	Young Adult - 62,143
Loyal customers	3,116

Prepared as a revised and professionally formatted version of the supplied project report summary.